BIOLOGICAL DATASET SELECTION

SWYNEX Internship — Task 1

Selected dataset: NCBI GEO GSE10072

Gene expression signature of cigarette smoking and its role in lung adenocarcinoma development and survival

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Project type: Public dataset selection and documentation; no laboratory work or new analysis performed.

Primary source: https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE10072

1. Introduction

Biological datasets allow researchers and students to explore patterns in biological measurements without collecting new samples themselves. The NCBI Gene Expression Omnibus (GEO) is a public repository that stores functional genomics studies, including gene-expression experiments. This project selects and describes one GEO study relevant to molecular biology and cancer research.

2. Dataset profile

Field	Description
Dataset title	Gene expression signature of cigarette smoking and its role in lung adenocarcinoma development and s
GEO Series	GSE10072
Curated GEO DataSet	GDS3257
Organism	Homo sapiens (human)
Experiment type	Expression profiling by microarray
Platform	GPL96 — Affymetrix Human Genome U133A Array
Final processed profiles	107 total: 58 tumour and 49 non-tumour lung tissue samples
Raw archive size	Approximately 375.3 MB (GSE10072_RAW.tar, as listed on GEO)

Size note: 107 refers to the final processed expression profiles reported on the GEO page. The 375.3 MB figure refers to the separate raw-data archive. These are different measures of dataset size.

3. Source and accessibility

The dataset is publicly available through NCBI GEO. Its record describes the study, organism, experiment type, platform, sample count and downloadable data files. The raw archive is listed as a TAR file containing CEL files. This project links to the original record rather than redistributing the large raw archive.

Primary record: https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE10072

Curated DataSet: https://www.ncbi.nlm.nih.gov/gds/3257

4. Research question

Proposed question: How do gene-expression patterns differ between lung adenocarcinoma tissue and non-tumour lung tissue, and how might smoking status relate to these patterns?

This question is suitable for a future exploratory analysis because the study includes tumour and non-tumour tissue profiles and records smoking information. It is a proposed question for this project; no new statistical analysis was conducted here.

5. Biological significance

Gene expression describes how strongly genes are expressed in a biological sample. Comparing expression patterns between tissue groups can help researchers generate hypotheses about pathways associated with disease. Smoking status may be relevant when interpreting lung tissue data, but other variables can also influence expression. A dataset can suggest associations; by itself, it cannot prove that an exposure caused a particular molecular change.

6. Proposed next steps

• Review sample annotations and confirm the tumour/non-tumour labels.

• Check the platform annotation and the meaning of the expression values.

• Inspect data quality and missing values before comparing groups.

• Choose an appropriate statistical workflow for comparing expression profiles.

• Consider smoking status and other potential confounding variables.

• Describe uncertainty and limitations before making any biological interpretation.

7. Limitations and responsible interpretation

• The study uses microarray expression profiling, not whole-genome sequencing.

• The two tissue groups have different sample counts (58 versus 49).

• Smoking status and other participant characteristics may affect observed patterns.

• Results from one dataset may not generalise to every population or setting.

• This submission documents a dataset selection only; no new analysis or experimental result is claimed.

8. Conclusion

GSE10072 is a relevant public biological dataset for learning how gene-expression studies can be described and used to develop a research question. The GEO record provides the key metadata needed for a transparent dataset-selection task: source, organism, platform, final sample count and raw archive size. A future analysis could explore expression differences between tumour and non-tumour lung tissue while carefully considering smoking status and other limitations.

9. References

• NCBI Gene Expression Omnibus. GSE10072: Gene expression signature of cigarette smoking and its role in lung adenocarcinoma development and survival. https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE10072

• NCBI GEO DataSets. GDS3257: Cigarette smoking effect on lung adenocarcinoma. https://www.ncbi.nlm.nih.gov/gds/3257

• NCBI GEO. About GEO DataSets. https://www.ncbi.nlm.nih.gov/geo/info/datasets.html

• NCBI GEO. Download GEO data. https://www.ncbi.nlm.nih.gov/geo/info/download.html

Prepared as a dataset-selection and documentation report for SWYNEX Internship Task 1. Please verify both submission links in an incognito/private window before submitting.